ECC arithmetic instruction-set pass: result

Status: field-level candidate measured; no qualified wall-time result. The run report is at research/ecc_arithmetic_20261013/reports/run-20261013T1837Z/report.md. The raw paired data are at research/ecc_arithmetic_20261013/paired_mac_final.csv.

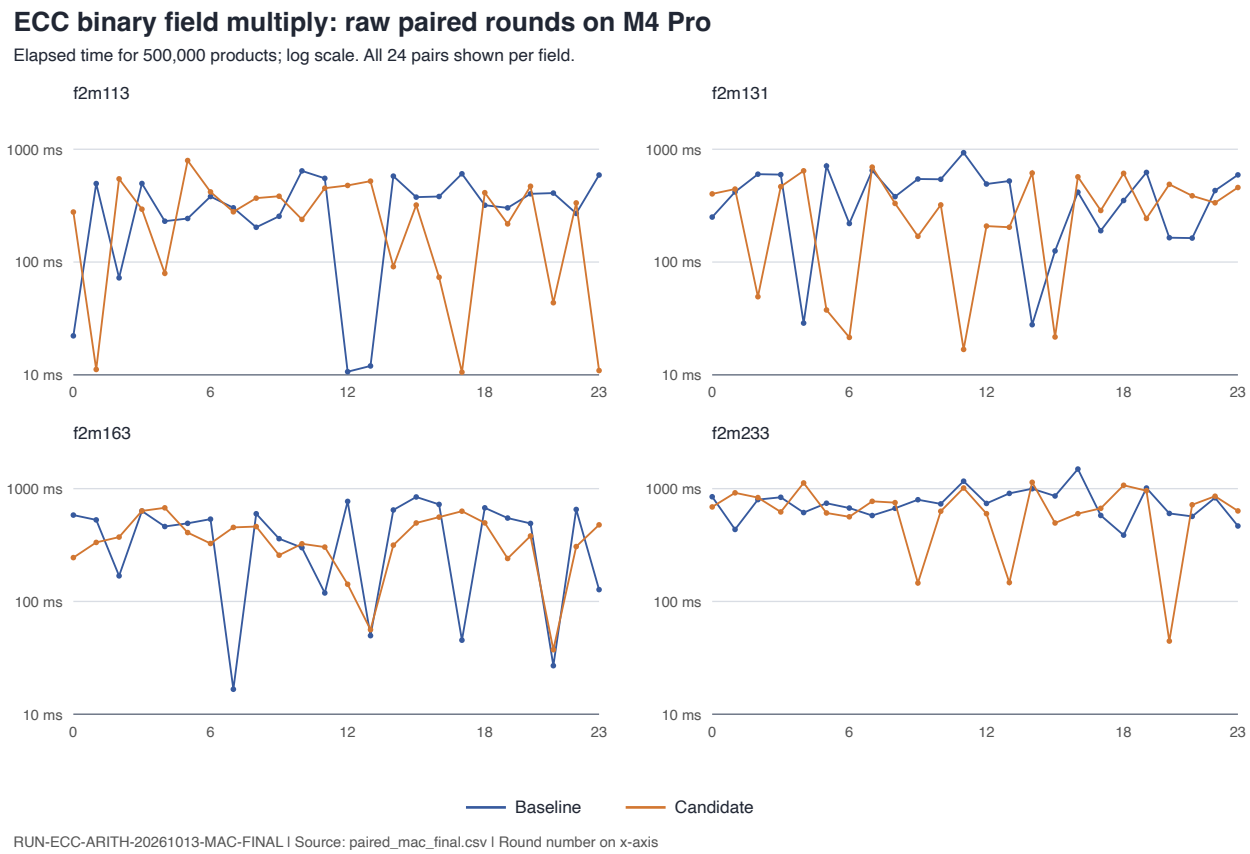
Implementation

The ARM PMULL and Intel PCLMUL paths now use three carryless products for two-limb elements and six for three-limb elements. The prior schoolbook path used four and nine. The four-limb and portable algorithms, reduction, public types and curve formulas remain the same. These are source-level operation counts, not a point or scalar performance result. The existing portable binary-field fallback has operand-dependent control flow and table access; this change does not validate a constant-time secret-scalar path.

The modified public multiply passed 17 focused release tests with reference equivalence and boundary fields. Both frozen sources compiled for x86-64 Linux with Rust 1.98.0; the later Modal panels executed them on x86-64. The full release library suite ran in the sandbox: 4,044 passed, nine local-socket tests failed with permission denied, and 114 were ignored. The collaboration group then passed 37/37 with loopback access. A complete loopback-enabled rerun was stopped under host load above 150, so a single full-suite pass is unverified.

Measurement decision

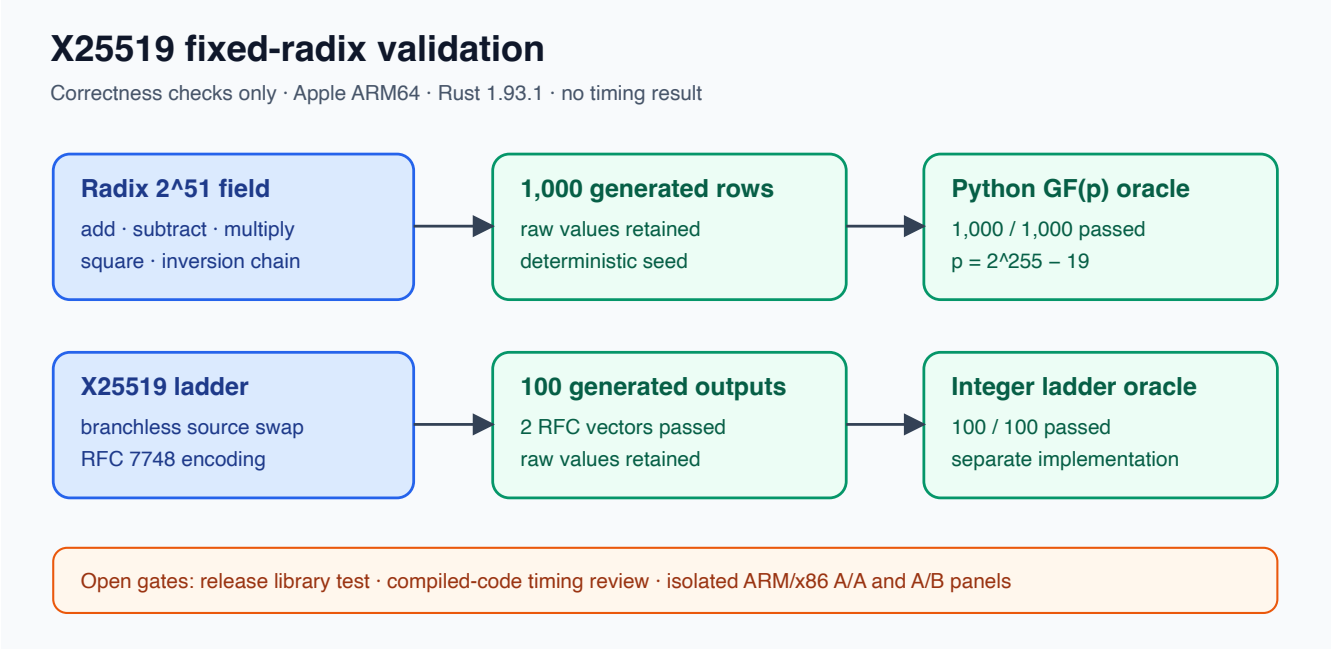
The final source-bound Apple M4 Pro panel completed 24 AB/BA pairs for each of GF(2^113), GF(2^131), GF(2^163), and GF(2^233), 500,000 multiplies per arm per round. All checksums matched. GF(2^131) baseline rounds ranged from 27,826,500 to 932,235,083 ns; candidate rounds ranged from 16,801,958 to 694,301,375 ns. The unchanged four-limb control also varied widely, and host load exceeded 100. No wall-time advantage or regression passes the controlled comparison gate.



After user authorization, two ephemeral Modal x86-64 workers completed the same frozen-source field panel: 34 tests passed per run, 24 AB/BA pairs per field, and 384 matching arm rows in total. The first launcher failed at a remote source path and its receipt remains. The completed workers reported PCLMULQDQ but no CPU model or reserved-core receipt; absolute times changed markedly between them, including the unchanged control. The dedicated Modal report and graph are in research/ecc_arithmetic_20261013/reports/modal-20261013/. No controlled wall-time or worldwide claim follows.

Next implementation: X25519

A five-limb radix-2^51 field and X25519 ladder were added after the field panels. Local source-bound checks passed 1,000 field rows against Python integers, 100 full X25519 outputs against an independent integer ladder, and two RFC 7748 vectors. The separate research/ecc_arithmetic_20261013/x25519_validation/REPORT.pdf shows the validation flow and receipt boundaries. A focused Cargo release build was stopped under host load before a test result; an x86-64 Modal export of the new source was rejected before launch pending specific authorization. No X25519 timing or machine-code constant-time conclusion follows. The smaller source-bound Cargo harness later passed 13 focused tests across the imported 25519 and P-521 modules; it does not replace the full suite.



Next implementation: P-521 field

A nine-word field modulo 2^521 - 1 is exposed in the ECC library. Its separate validation flow and raw receipt are in research/ecc_arithmetic_20261013/p521_validation/REPORT.pdf and README.md. The Rust source emitted 1,000 deterministic rows; a Python integer oracle matched canonicalization, addition, subtraction, multiplication and inversion in every row. No P-521 point or scalar path uses the field, and no timing or machine-code side-channel conclusion follows. P-192/224/384, Ed25519, Koblitz recoding, Metal and CUDA/HIP remain open.

